Submission 1

2026-10-21

Step 1: Clean the covariate data

```
head(GSE.covar.raw)
```

```
## # A tibble: 6 x 10
##   subjectID geoID      mat_age_yrs        preg_preeclampsia delivery_gestAge_wks
##   <chr>     <chr>      <chr>              <chr>             <chr>
## 1 6053887-M GSM2264346 age_years: 31.066~ pregnancy_condit~ gestation_weeks: 37~
## 2 3064425-M GSM2264347 age_years: 27.794~ pregnancy_condit~ gestation_weeks: 39~
## 3 6094898-M GSM2264348 age_years: 34.318~ pregnancy_condit~ gestation_weeks: 38~
## 4 7050617-M GSM2264349 age_years: 33.426~ pregnancy_condit~ gestation_weeks: 39~
## 5 7016015-M GSM2264350 age_years: 28.279~ pregnancy_condit~ gestation_weeks: 40~
## 6 7054360-M GSM2264351 age_years: 23.501~ pregnancy_condit~ gestation_weeks: 41~
## # i 5 more variables: sampleCollection_gestAge_wks <chr>,
## #   batch_Microarray <chr>, mat_BMI <chr>, mat_vitD <chr>, mat_asthma <chr>
```

```
summary(GSE.covar.raw)
```

```
##      subjectID         geoID        mat_age_yrs  preg_preeclampsia
##  Length   :157   Length   :157   Length   :157   Length   :157
##  N.unique :157   N.unique :157   N.unique :157   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  9   Min.nchar: 10   Min.nchar: 13   Min.nchar: 27
##  Max.nchar:  9   Max.nchar: 10   Max.nchar: 22   Max.nchar: 33
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Length   :157        Length   :157                 Length   :157
##  N.unique : 44        N.unique : 53                 N.unique :  8
##  N.blank  :  0        N.blank  :  0                 N.blank  :  0
##  Min.nchar: 19        Min.nchar: 32                 Min.nchar: 41
##  Max.nchar: 28        Max.nchar: 41                 Max.nchar: 41
##       mat_BMI         mat_vitD       mat_asthma
##  Length   :157   Length   :157   Length   :157
##  N.unique :140   N.unique :132   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar: 20   Min.nchar: 45   Min.nchar: 31
##  Max.nchar: 22   Max.nchar: 48   Max.nchar: 31
```

```
# Data Wrangling time! Whoohoo! Seems like there are some extra text in some
# columns that make unwanted 'noise'. With the help of the trusty mutate function
# we can clean these cells up!
GSE.covar <- GSE.covar.raw %>%
```

```
  mutate(
    mat_age_yrs = str_remove_all(mat_age_yrs,"age_years: "),
    preg_preeclampsia = str_remove_all(preg_preeclampsia, "pregnancy_condition: "),
    delivery_gestAge_wks = str_remove_all(delivery_gestAge_wks, "gestation_weeks: "),
    sampleCollection_gestAge_wks = str_remove_all(sampleCollection_gestAge_wks, "blood_draw_at_gestatio
    batch_Microarray = str_remove_all(batch_Microarray, "microarray_hybridization_date: "),
    mat_BMI = str_remove_all(mat_BMI, "bmi_kg/m2: "),
    mat_vitD = str_remove_all(mat_vitD, "vitamind_25\\(0h\\)_ng/ml_whole_blood_baseline: "),
    mat_asthma = ifelse(str_ends(mat_asthma, "1"), "1", "0"))


# I also noticed all varaibles are stored as characters, for safe keeping
# I am going to change the ones that look like numeric measure to numeric
GSE.covar <- GSE.covar %>%
  mutate(across(c(mat_age_yrs, delivery_gestAge_wks, sampleCollection_gestAge_wks,
                  mat_BMI, mat_vitD),
                as.numeric))


head(GSE.covar)
```

```
## # A tibble: 6 x 10
##   subjectID geoID      mat_age_yrs preg_preeclampsia delivery_gestAge_wks
##   <chr>     <chr>            <dbl> <chr>                            <dbl>
## 1 6053887-M GSM2264346        31.1 Preeclampsia                      37.4
## 2 3064425-M GSM2264347        27.8 Normal                            39.4
## 3 6094898-M GSM2264348        34.3 Normal                            38.9
## 4 7050617-M GSM2264349        33.4 Normal                            39.1
## 5 7016015-M GSM2264350        28.3 Preeclampsia                      40.1
## 6 7054360-M GSM2264351        23.5 Normal                            41.1
## # i 5 more variables: sampleCollection_gestAge_wks <dbl>,
## #   batch_Microarray <chr>, mat_BMI <dbl>, mat_vitD <dbl>, mat_asthma <chr>
```

```
dim(GSE.covar)
```

```
## [1] 157  10
```

Step 2: Make the categorical variables

```
# Categorical variable #1: preeclampsia status (already categories, so we just
# tell R it's a factor and put "Normal" first so it is the reference group).
# Categorical variable #2: vitamin D status, DERIVED from the continuous
# mat_vitD column. Under 20 ng/mL is a commonly used cutoff for deficiency.

GSE.covar <- GSE.covar %>%
  mutate(
    preg_preeclampsia = factor(preg_preeclampsia,
                               levels = c("Normal", "Preeclampsia")),
    vitD_status = factor(ifelse(mat_vitD < 20,
                                "Deficient (<20 ng/mL)",
```

```
                                "Sufficient (>=20 ng/mL)"),
                         levels = c("Deficient (<20 ng/mL)",
                                    "Sufficient (>=20 ng/mL)")))


# Check that the new variable makes sense: group sizes, and that the cutoff
# really split the vitamin D values where we said it would.
table(GSE.covar$preg_preeclampsia, GSE.covar$vitD_status)
```

```
##
##                 Deficient (<20 ng/mL) Sufficient (>=20 ng/mL)
##   Normal                           52                      58
##   Preeclampsia                     26                      21
```

```
GSE.covar %>% group_by(vitD_status) %>% summarise(min = min(mat_vitD), max = max(mat_vitD))
```

```
## # A tibble: 2 x 3
##   vitD_status               min   max
##   <fct>                   <dbl> <dbl>
## 1 Deficient (<20 ng/mL)    5.11  19.9
## 2 Sufficient (>=20 ng/mL) 20.4   80.8
```

Step 3: Clean the gene expression data and attach gene names

```
# The first column of the expression file has a blank header, so we name it
# PROBEID. Probe IDs are just labels, so we store them as text in both tables
# so that R can match them up.

GSE.GE <- GSE.GE.raw %>%
  rename(PROBEID = 1) %>%
  mutate(PROBEID = as.character(PROBEID))

GSE.GN <- GSE.GN.raw %>%
  mutate(PROBEID = as.character(PROBEID))

# Step A: look up the gene name for each probe
# Step B: pivot_longer turns the sideways table (one column per person) into a
#         tall table with one row per gene per person.

GSE.GE <- GSE.GE %>%
  left_join(GSE.GN, by = "PROBEID") %>%
  pivot_longer(cols = -c(PROBEID, GENE),
               names_to = "geoID",
               values_to = "expression")

# Safety checks: every probe got a gene name, and the sample IDs in the
# expression data are exactly the sample IDs in the covariate data.
stopifnot(!any(is.na(GSE.GE$GENE)))
stopifnot(setequal(unique(GSE.GE$geoID), GSE.covar$geoID))

head(GSE.GE)
```

```
## # A tibble: 6 x 4
##   PROBEID GENE    geoID      expression
##   <chr>   <chr>   <chr>           <dbl>
## 1 8089659 CCDC191 GSM2264363       6.18
## 2 8089659 CCDC191 GSM2264379       6.11
## 3 8089659 CCDC191 GSM2264394       6.32
## 4 8089659 CCDC191 GSM2264418       5.91
## 5 8089659 CCDC191 GSM2264449       6.35
## 6 8089659 CCDC191 GSM2264452       5.91
```

Step 4: Pick our gene and build the plotting table

```
# Not every gene in the linker file is in the expression data, so check first!
gene_of_interest <- "RUNDC3A"
gene_of_interest %in% GSE.GE$GENE   # should be TRUE
```

```
## [1] TRUE
```

```
# Keep just our gene, then attach each person's covariates using geoID
dat <- GSE.GE %>%
  filter(GENE == gene_of_interest) %>%
  left_join(GSE.covar, by = "geoID")

# Should be 157 rows: one per person
stopifnot(nrow(dat) == nrow(GSE.covar))
head(dat)
```

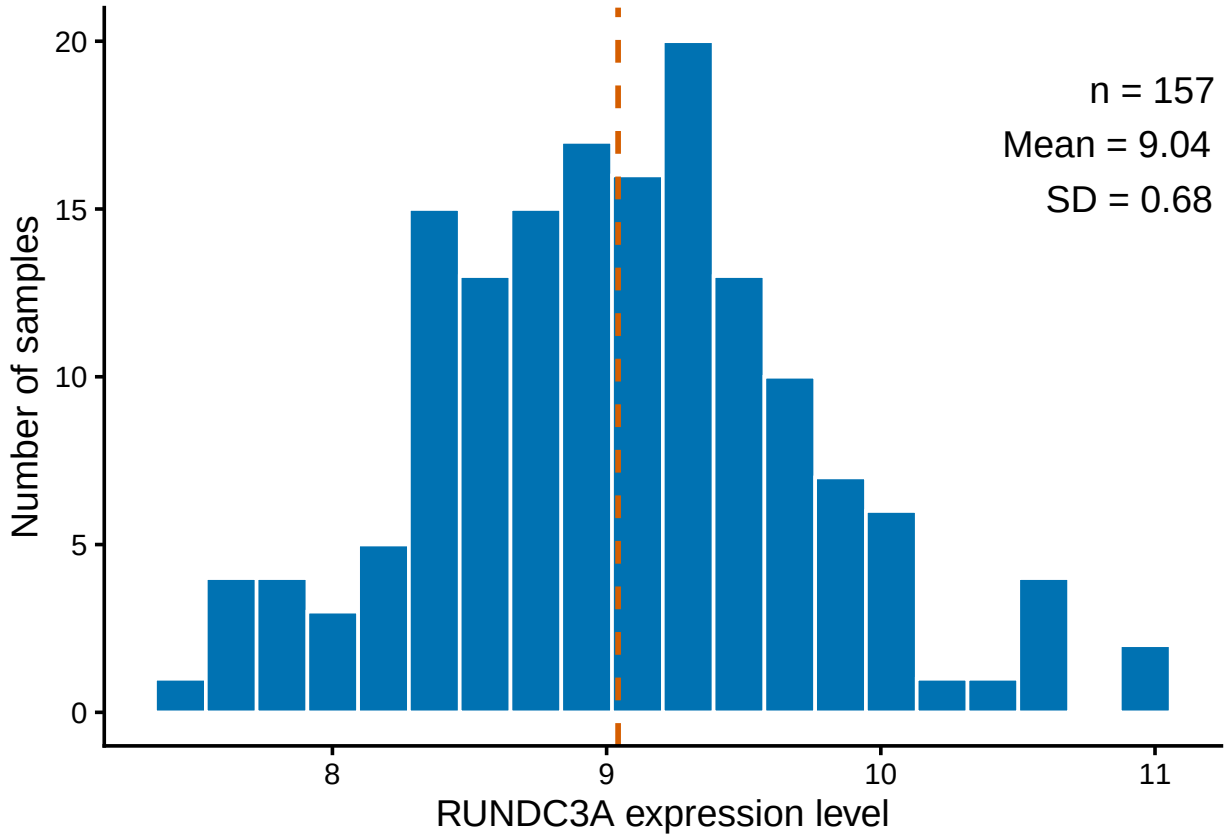
```
## # A tibble: 6 x 14
##   PROBEID GENE    geoID      expression subjectID mat_age_yrs preg_preeclampsia
##   <chr>   <chr>   <chr>           <dbl> <chr>           <dbl> <fct>
## 1 8007607 RUNDC3A GSM2264363       9.16 8054547-M        33.1 Normal
## 2 8007607 RUNDC3A GSM2264379       8.61 6051478-M        26.6 Normal
## 3 8007607 RUNDC3A GSM2264394       8.47 1090275-M        28.4 Normal
## 4 8007607 RUNDC3A GSM2264418       8.68 1075381-M        20.9 Normal
## 5 8007607 RUNDC3A GSM2264449       9.43 1084308-M        18.9 Preeclampsia
## 6 8007607 RUNDC3A GSM2264452       7.70 6017807-M        20.9 Normal
## # i 7 more variables: delivery_gestAge_wks <dbl>,
## #   sampleCollection_gestAge_wks <dbl>, batch_Microarray <chr>, mat_BMI <dbl>,
## #   mat_vitD <dbl>, mat_asthma <chr>, vitD_status <fct>
```

Plot 1: Histogram of gene expression

```
m <- mean(dat$expression); s <- sd(dat$expression)

ggplot(dat, aes(x = expression)) +
  geom_histogram(bins = 20, fill = "#0072B2", color = "white") +
   geom_vline(xintercept = m, color = "#D55E00", linewidth = 1, linetype = "dashed") +
  annotate("text", x = Inf, y = Inf, hjust = 1.05, vjust = 1.5, size = 5,
           label = sprintf("n = %d\nMean = %.2f\nSD = %.2f", nrow(dat), m, s)) +
```

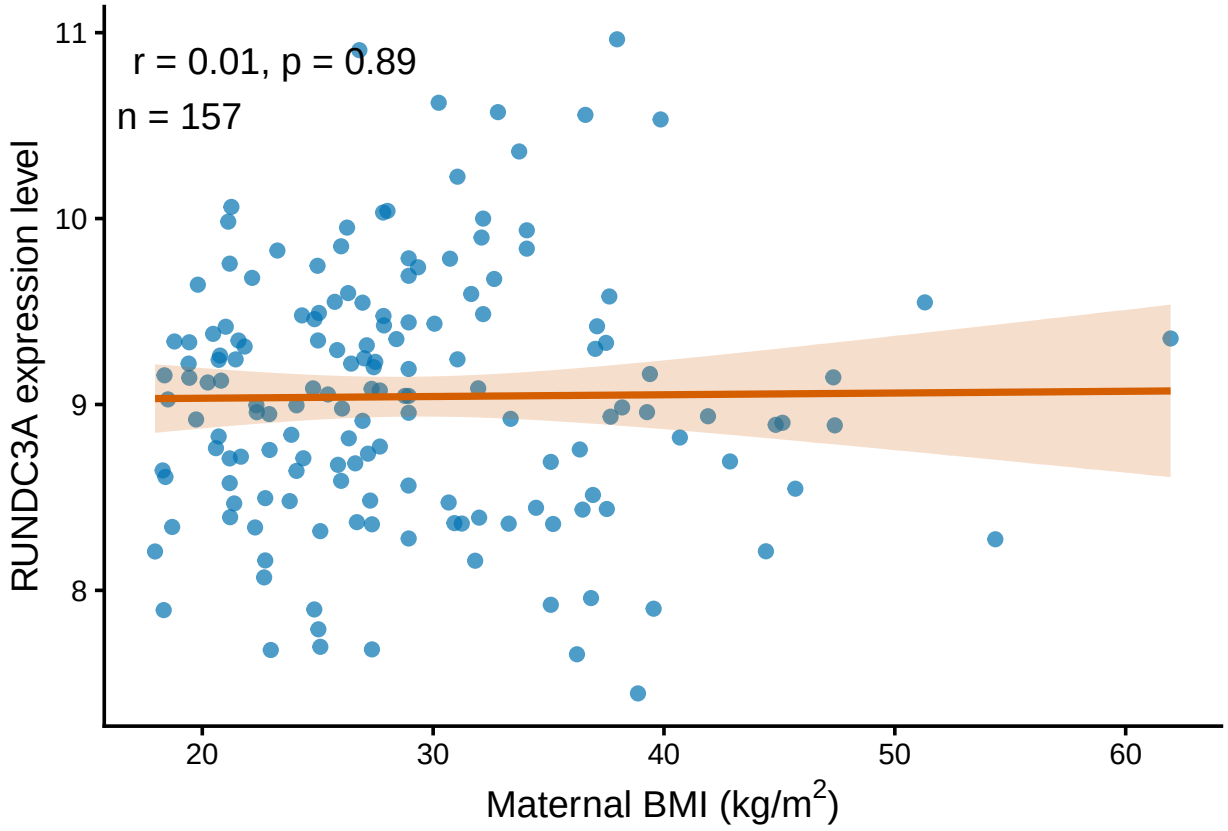
```
  labs(x = paste(gene_of_interest, "expression level"),
       y = "Number of samples") +
  theme_classic(base_size = 14)
```



Plot 2: Scatterplot of covariate vs. gene expression

```
ct <- cor.test(dat$mat_BMI, dat$expression)

# Each dot is one person. The line is a straight-line fit with its
# 95% confidence band, to show the overall trend.
ggplot(dat, aes(x = mat_BMI, y = expression)) +
  geom_point(alpha = 0.7, size = 2, color = "#0072B2") +
  geom_smooth(method = "lm", formula = y ~ x, color = "#D55E00",
              fill = "#D55E00", alpha = 0.2) +
  annotate("text", x = -Inf, y = Inf, hjust = -0.1, vjust = 1.5, size = 5,
           label = sprintf("r = %.2f, p = %.2f\nn = %d", ct$estimate, ct$p.value, sum(!is.na(dat$mat_BMI
  labs(x = expression("Maternal BMI (kg/m"^2*")"),
       y = paste(gene_of_interest, "expression level")) +
  theme_classic(base_size = 14)
```



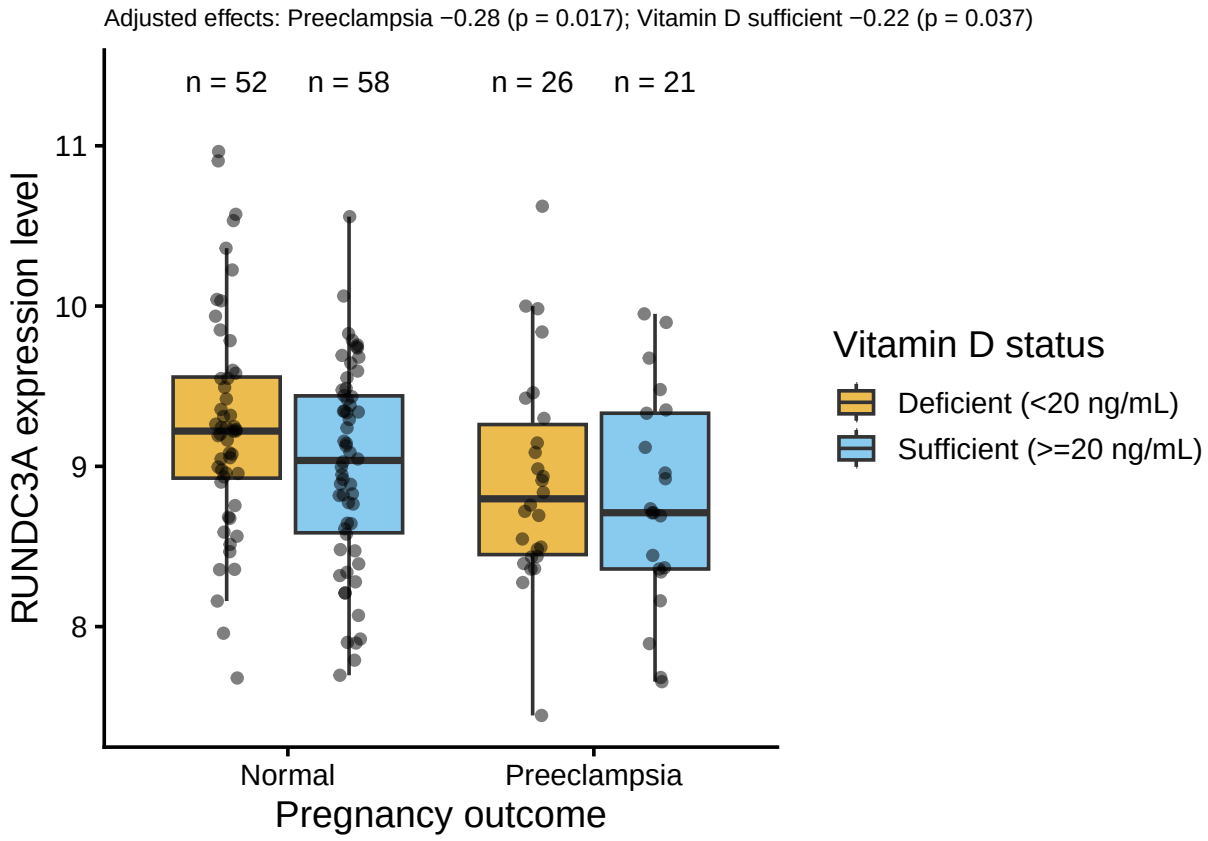
Plot 3: Boxplot of gene expression by both categorical variables

```
fit <- lm(expression ~ preg_preeclampsia + vitD_status, data = dat)
coef <- summary(fit)$coefficients

sub_text <- sprintf(
  "Adjusted effects: Preeclampsia %.2f (p = %.3f); Vitamin D sufficient %.2f (p = %.3f)",
  coef[2, "Estimate"], coef[2, "Pr(>|t|)"],
  coef[3, "Estimate"], coef[3, "Pr(>|t|)"]
)

# x-axis = preeclampsia status, colour = vitamin D status.
# Dots show the individual samples so readers can see the group sizes.
ggplot(dat, aes(x = preg_preeclampsia, y = expression, fill = vitD_status)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7,
               position = position_dodge(width = 0.8), width = 0.7) +
  geom_point(aes(group = vitD_status),
             position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.8),
             alpha = 0.5, size = 1.5, show.legend = FALSE) +
  stat_summary(fun.data = function(y) data.frame(y = 11.4, label = paste0("n = ", length(y))),
               geom = "text", position = position_dodge(width = 0.8), size = 4) +
  scale_fill_manual(values = c("#E69F00", "#56B4E9")) +
  labs(x = "Pregnancy outcome",
```

```
      y = paste(gene_of_interest, "expression level"),
      fill = "Vitamin D status",
      subtitle = sub_text) +
 theme_classic(base_size = 14) +
 theme(plot.subtitle = element_text(size = 9))
```



ADDITIONAL RESOURCES USED:

- Google & Gemini: to look up commands we learned in class, and extra commands for more advanced arguments.
- Claude (sonnet, medium): to check my work for errors, and help me 'dumb down'/understand specific characteristics in the data and rstudio itself.